USN | | | | | | | | | | [22MTDSDE52]

THIRD SEMESTER M.Tech EXAMINATION, DECEMBER - 2024

SOCIAL NETWORK AND MINING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Justify different Types of Social Networks and write in detail.	CO1	L4	**10**
	b.	Elaborate any two detail Techniques for Data collection.	CO1	L3	**10**
2.	**a.**	Analyze different ways to represent graph structures.	CO2	L4	**10**
	b.	Evaluate the effectiveness of different web scraping tools and techniques.	CO2	L5	**10**
3.	**a.**	What is modularity in the context of community detection? Apply the modularity-based methods to detect communities in a network and why community detection is important.	CO3	L3	**10**
	b.	Describe the use of network visualization techniques. How do these techniques help in exploring and interpreting the structure of networks? Provide an example.	CO3	L3	**10**
		OR			
4.	**a.**	Discuss the small-world phenomenon and scale-free networks. What are the key characteristics of each, and how do they differ from each other?	CO3	L3	**20**
5.	**a.**	Discuss the process of influence analysis in social networks, including identifying influential nodes and understanding how information diffusion models work. How can these models be used for viral marketing?	CO4	L3	**20**
		OR			
6.	**a.**	Explain how machine learning approaches can be applied to link prediction in social networks. What are the advantages of using these methods compared to similarity-based approaches?	CO4	L2	**10**
	b.	Describe the role of opinion mining and sentiment analysis in social networks. How do these techniques help understand the user behavior and sentiment trends?	CO4	L2	**10**

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7.	**a.**	What are the ethical and privacy issues associated with social network analysis and mining? Why are these concerns important, and what are some strategies to address them?	CO5	L2	**10**
	b.	What are temporal networks, and how do they differ from static networks? Why are the analysis of temporal networks important for understanding real-world systems?	CO6	L2	**10**
		OR			
8.	**a.**	Explain the key concepts of dynamic network analysis, including temporal networks, event detection, and evolution analysis. How are these techniques used to understand the changing nature of networks over time?	CO5	L3	**20**